

Figure 4D. Top model comparison table

Model	Family	ROC-AUC	PR-AUC	F1	Accuracy	Brier
MegaWeighted HGT+EdgeAware+HeteroSAG	Final ensemble	0.872	0.920	0.833	0.796	0.149
weighted_graph0.6_ xgb0.2_nn0.2	Graph learning	0.832	0.890	0.810	0.765	0.165
GraphSAGE	Graph learning	0.831	0.889	0.811	0.763	0.165
weighted_graph0.5_ xgb0.25_nn0.25	Graph learning	0.829	0.888	0.807	0.760	0.166
weighted_graph0.7_ nn0.3	Graph learning	0.831	0.888	0.805	0.761	0.166
weighted_graph0.6_ nn0.4	Graph learning	0.829	0.887	0.807	0.763	0.167
weighted_graph0.7_ xgb0.3	Graph learning	0.828	0.886	0.806	0.758	0.168
weighted_graph0.8_ xgb0.2	Graph learning	0.828	0.885	0.802	0.755	0.169
weighted_graph0.4_ xgb0.3_nn0.3	Graph learning	0.824	0.885	0.799	0.749	0.168
weighted_graph0.6_ xgb0.4	Graph learning	0.826	0.885	0.809	0.760	0.168